A1 Week 2 Plan

Role 1 (Optimal Contract and General Carbon Theory)

- **Establish existence, uniqueness, and the optimal contract.** Prove coercivity of expected profit and a primitive condition on F (e.g. a log-concave density) under which the interior capacity optimum is unique, and prove that the dispatch contract $d^*(x)=\min\{(K_F-x)_+, K_M\}$ is the unique optimal policy via the strict $R>c$ ordering. This discharges the currently-assumed interior optimum and supplies the negative-definite-Hessian condition from primitives.
- **Promote the shift-ratio condition to a primitive.** Replace the endogenous-window condition $g(x)=F_{K_R}/f$ non-increasing with a support-wide condition on F that implies it, and prove the implication.
- **Prove the necessary-and-sufficient emission condition for general F.** Under that primitive plus a tail/skew condition, prove a sharp iff for carbon reduction at a finite interior optimum, upgrading the endogenous inequality (C) and the current existence conjecture to a clean condition in primitives. Exhibit a non-power-law F satisfying all three conditions to certify the class is non-vacuous and genuinely more general than the power-law/uniform case.
- **Complete the viability threshold.** Extend the renewable-cost window from the power-law family to general F, with the strict fall at a finite interior K_R^* rather than the unbounded corner, and prove the carbon-reduction set is an interval (single-crossing in C_R) so carbon falls if and only if C_R is below the threshold.
- **State the main theorem and re-scope.** State the headline theorem, align the abstract and title to exactly what is proved, and write the economic intuition at journal quality.

Role 2 (Solver and Numerical Methods)

- **Build and document the general solver.** Finalize the joint-capacity and dispatch solver for $d^*(x)$ and the capacities under any residual-demand distribution — the validated backend the rest of the group relies on.
- **Deliver one carbon-sign narrative.** Replace the scattered counts with a principled sweep over Role 1's primitive condition on a curated distribution family, showing carbon falls at a finite interior optimum.
- **Provide a Hessian / existence certificate.** A reproducible check that the Hessian is negative definite on the interior path for Role 1's distribution class, backstopping existence and uniqueness.
- **Harden the solver and ship reproducibility.** Replace finite-difference cofactors with analytic ones to remove the noisy knife-edge points, and ship a single command that reproduces the exact-formula checks and the carbon sweep, with documented tolerances.

Role 3 (Comparative Statics and Robustness)

- **Prove the correlation and price comparative statics.** Show the net-carbon sign is governed by the demand–supply association (the shift ratio $g = E[S \mid x]$); that mining raises optimal renewable capacity $K_R^* > K_R^\circ$; and sign the monotonicities in the computing margin R, the blackout penalty β, and the retail price p, under general conditions.
- **Establish the cross-distribution robust certificate.** Sign the capacity covariation along the shape class U so the least-favorable result holds across F, making the renewable-cost condition a genuine distributionally-robust certificate, or prove the obstruction cleanly.
- **Construct the reversed-channel counterexample.** Build a realized F with g increasing so that $K_R^* < K_R^\circ$ — the sharpness example Week 1 could not produce, since an algebraic sign-flip is not a realized example.
- **Repair the limit-interchange.** Replace the expected-fossil-to-zero corner step with a dominated-convergence or finite-capacity envelope argument.

Role 4 (Mechanism Design, Regimes, and Positioning)

- **Strengthen the mechanism-design treatment.** Cast the regulator as principal and the utility/miner as agent; write the participation and incentive-compatibility constraints and prove the price-based demand-response program implements d^* optimally and, where feasible, uniquely — not merely as the principal's first-best dispatch; state the conditions (price-taking, exogenous R) under which those constraints are slack and the induced miner-demand result holds.
- **Unify and complete the regime map.** Use one Assumption 1 and one carbon decomposition throughout; fold the finite-optimum statement and the renewable-cost boundedness axis into a single consistent regime theorem with the Assumption-1 necessity pattern.
- **State the limitations and scope boundary.** State explicitly what is assumed and what is deferred to the companion workstreams (calibration; reliability, welfare, and pricing; the flexibility continuum and dynamics), so no referee can reject on an unaddressed omission rather than an error in scope.
- **Position and assemble.** Write the related-work section on the methods used — monotone comparative statics, interruptible and demand-response load economics, and distributionally-robust analysis — add the bibliography, and assemble the pieces into one manuscript with a single notation table.